

Aarti Darji

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EDUCATION

M.S. Computer Science, *New York University, Courant Institute* Jan 2026-Dec 2027
Honors B.S. Data Science, *University of Texas at Arlington*, 3.65/4.00 Aug 2021-May 2025

SKILLS

Programming Languages Python, R, SAS, MATLAB, SQL, JavaScript, C++, HTML/CSS
Libraries/Frameworks React.js, Node.js, Flask, Pandas, Matplotlib, Seaborn, TensorFlow, Pytorch
Lightning, CUDA, Scanpy, OpenCV, ggplot, tidyverse.
Tools Git, Github, Linux, Bash, Slurm, ImageJ, QuPath, Figma, Microsoft Office, FSL, FreeSurfer.
Relevant Coursework Python for Data Science 1 & 2, Data Mining, Machine Learning, Statistical Methods 1 & 2, Statistical Inference, Behavioral Genetics.

RESEARCH EXPERIENCE

Research Assistant I - Analyzing Gulf War Illness in Veterans June 2025- Present
Advisor: Dr. Deborah Little *UTHealth Houston*

- Curated and maintained raw and processed MRI data for Gulf War Illness studies in veterans.
- Performed diffusion MRI analysis using FSL and FreeSurfer for structural and tract-based insights.

Undergraduate Research Intern - Agentic AI for Medical Imaging June 2024- May 2025
Advisor: Dr. Nils Gehlenborg *Harvard Medical School*

- Developed a chatbot interface for Vitesse, a visual integration tool, that facilitated natural language exploration for multiplexed immunofluorescence (mIF) datasets.
- Create a multi-agent framework that supported biomarker exploration within the tumor microenvironment (TME) and location of regions of interests (ROIs).
- Implemented few shot learning and a RAG approach for accurate information retrieval and obtaining structured output.

Undergraduate Research Assistant - SSL for Medical Imaging May 2023 - May 2024
Advisor: Dr. Jacob Luber *Health Data Science Lab*

- Created an integrated search framework allowing retrieval of most similar spatial proteomics (CODEX/PhenoCycler) imaging from input histopathology slide.
- Developed and trained a Variational Autoencoder (VAE) on HPC clusters to obtain latent spaces of patches to perform similarity search.
- Conducted validation and ablation studies displaying potential of search engines to aid clinicians in making diagnoses and deciding treatment outcomes.

Undergraduate Research Assistant - Hybrid Design and Fabrication Jan 2023 - May 2024
Advisor: Dr. Cesar Torres *Hybrid Atelier*

- Lasercut and 3D-Printed molds for developing silicone samples with different micro-textures to enhance haptic perception and study the effect on emotions and decision making.
- Created a web application using React and Flask API to collect user responses for study and observe texture perception of silicone samples.
- Developed a layered silicone composite sensor with unique electrode designs to study new gesture creation and cross-modal correspondences.
- Achieved 97% accuracy by training a CNN model to predict user gestures based on sensor reading heatmaps.

Undergraduate Research Assistant - Cognitive Science Aug 2021- June 2022
Advisor: Dr. Matthew Robison and Dr. Pedro Maia *Controlled Attention and Memory Lab*

- Developed python code to provide useful insight on average reaction time and pupil diameter/gaze shift obtained from eye-tracking devices for different conditions for 300+ participants.

- Calculated the sample entropy of pupil diameter to determine the regularity of pupil diameter based on type of goal, feedback, and reward.

PUBLICATIONS

- [1] Amir Hajighasemi*, M. D. Saurav*, **Aarti Darji***, Mohammad S. Nasr, Jai Prakash Veerla, Parisa Boodaghi Malidarreh, Michael Robben, Helen H. Shang, and Jacob M. Lubner. “*Multimodal Pathology Image Search Between H&E Slides and Multiplexed Immunofluorescent Images.*” 2024 IEEE Conference on Computational Intelligence in Bioinformatics and Computational Biology (CIBCB). IEEE, 2024. (Short Paper)
- [2] **Darji, Aarti**, Gunnika Kapoor, and Cesar Torres. ‘*Trace to Touch: Eliciting Gestures from Capacitive Touch Electrodes.*’ Proceedings of the 16th Conference on Creativity & Cognition 2024. (Extended Abstract – Short Paper)
- [3] Kapoor, Gunnika, **Aarti Darji**, Marisa N. Fernandez, and Cesar Torres. ‘*Sentura: Understanding the Cognitive Affordances of Silicone Microtextures in Tangible User Interface Design.*’ In Proceedings of the 16th Conference on Creativity & Cognition. (Extended Abstract – Short Paper)
- [4] Veerla, Jai Prakash, Partha Sai Guttikonda, Amir Hajighasemi, Jillur Rahman Saurav, **Aarti Darji**, Cody T. Reynolds, Mohamed Mohamed, Mohammad S. Nasr, Helen H. Shang, and Jacob M. Lubner. SpatialVisVR: An Immersive, Multiplexed Medical Image Viewer With Contextual Similar-Patient Search. 2024 IEEE Conference on Computational Intelligence in Bioinformatics and Computational Biology (CIBCB). IEEE, 2024.

POSTER & PRESENTATIONS

- [1] **Darji, Aarti***, Amir Hajighasemi*, M. D. Saurav*, Mohammad S. Nasr, Jai Prakash Veerla, Parisa Boodaghi Malidarreh, Michael Robben, Helen H. Shang, and Jacob M. Lubner. *Integrated Search Framework for Histopathology: From H&E to Multiplexed Immunofluorescence.* Intelligent Systems of Molecular Biology (ISMB) 2024 (Poster).
- [2] **Aarti Darji**, Eric Moerth, Morgan Turner, David Kouril, Jacob Lubner, Nils Gehlenborg. *LLM-Supported Exploration and Analysis of 3D-Microscopy Imaging.* Bio+MedVis Challenge @ IEEE VIS 2024 (Talk & Poster).
- [3] **Aarti Darji**, Eric Mörth, Morgan Turner, David Kouřil, Nils Gehlenborg. *Ask and Reveal: Natural Language-Driven Analysis of Multiplexed Immunofluorescence Imaging.* Intelligent Systems of Molecular Biology (ISMB) 2025 (Poster).

OTHER PUBLICATIONS

- [1] **Darji, Aarti.** "Revolutionizing Cancer Research with AI: Health Data Science Lab, UT Arlington." *XRDS: Crossroads, The ACM Magazine for Students* 30, no. 4 (2024): 64-65.

AWARDS & SCHOLARSHIPS

Maverick Academic Scholarship 2021-2025
 Honors Residential Learning Community Housing Grant 2021-2022
 College of Science Dean’s Leadership Circle Award 2022-2023
 College of Science Dean’s Leadership Circle Award 2023-2024
 SWPA Data Science in Psychology Travel Grant 2023
 OurCS@DFW 2023 Best Presentation Award
 TExBioMed Summer Research Program 2023 Outstanding Presentation Award
 Honors College Carolyn A. Barros Founding Dean Scholarship 2024-2025

LEADERSHIP

Vice President, <i>Psychology Society</i>	Dec 2022 – May 2023
Research Director, <i>ACM@UTA</i>	Sep 2022 – May 2023
Programming Board Coordinator, <i>Honors College</i>	Sep 2022 – May 2023
Veneratio Newsletter Staff, <i>Honors College</i>	Sep 2022 - Dec 2022